Forecasting-Based Bias Correction in Difference-in-Differences

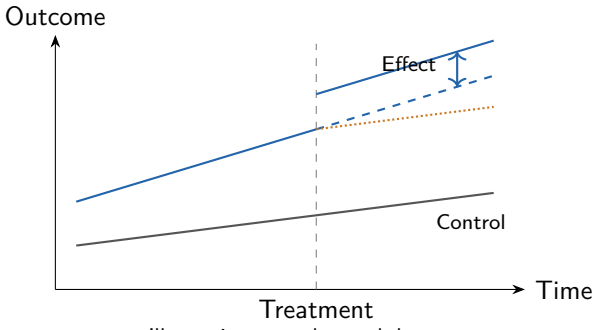
Simulation and Empirical Evidence

Hongqiang Shi

Supervisor: Klaus Ackermann

Master of Applied Econometrics
Department of Econometrics and Business Statistics
Monash University

Differential Untreated Trends in DID



Illustration, not observed data

Solid blue: observed treated-group outcomes.
Dashed blue: treated-group path without treatment.
Dotted orange: counterfactual under parallel trends.

- Fixed effects remove persistent levels and common shocks.
- Differential untreated growth remains.
- DID combines the treatment effect with the untreated gap change.

Research Questions and Study Design

1. When can forecasting improve DID estimates?
2. How should we evaluate these methods?
3. How can we choose a method when the true effect is unknown?

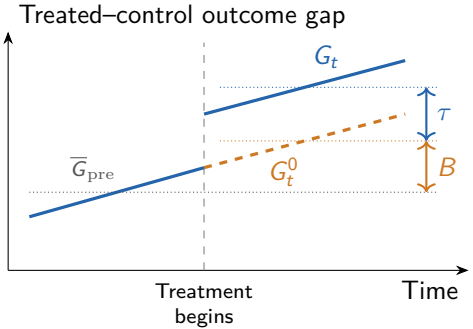
Simulation: we know the true effect

Set the true effect to 2, create different trend violations, and check how close each estimate gets to the truth.

Ceará: we do not know the true effect

Test the methods on earlier, pre-strike periods, choose using those results, and then estimate the strike-period contrast.

A Common Gap-Change Correction



Conceptual illustration. Horizontal guides show period means.

G_t: observed gap between the two groups.

G_t^0: gap that would exist without treatment.

$$\widehat{\tau}_{\text{TWFE}} = \overline{G}_{\text{post}} - \overline{G}_{\text{pre}},$$

$$\widehat{B}_m = \overline{\widehat{G}}^{\,0}_{m,\text{post}} - \overline{G}_{\text{pre}},$$

$$\widehat{\tau}_{\text{adj},m} = \widehat{\tau}_{\text{TWFE}} - \widehat{B}_m = \overline{G}_{\text{post}} - \overline{\widehat{G}}^{\,0}_{m,\text{post}}.$$

1. Measure the observed change in the group gap.
2. Forecast how the gap would have changed without treatment.
3. Subtract that predicted change from DID.

Candidate Forecasting Procedures

Estimator	Forecasting procedure
TWFE	Common two-way fixed effects benchmark.
Matched gap adjustment (Oracle)	Uses the correct trend form for each simulation scenario; coefficients are estimated from pre-treatment data.
Linear adj.	Common slope; predicted gap equals the pre-period mean gap.
SC adj.	Synthetic Control; single-mean matching with contemporaneous donors.
RF adj.	Random Forest; aggregate both groups' outcome predictions.
GB adj.	Gradient Boosting; aggregate both groups' outcome predictions.

Oracle knows the functional form, not the true coefficients or future outcomes.
The empirical Linear gap differs from the simulation's common-slope Linear.

Creating Parallel-Trend Violations

$$
\begin{aligned}
Y_{it}(0) &= \alpha_i + 0.10t + \theta_i f_s(t) + \varepsilon_{it}, \\
\Pr(D_i = 1 \mid \theta_i) &= \text{logit}^{-1}(-0.40 + \delta_{1,s}\theta_i), \\
Y_{it} &= Y_{it}(0) + 2D_i\,\mathbf{1}\{t > 12\}.
\end{aligned}
$$

1. **Units have different untreated growth rates.**
   θ_i controls how strongly each unit follows $f_s(t)$.
2. **Faster-growing units are more likely to be treated.**
   Treatment probability increases with θ_i in violation scenarios.
3. **The untreated group gap changes over time.**
   DID picks up this gap change as well as the treatment effect.

Implemented Simulation Scenarios

Scenario	Trend function and description	$\sigma_{\theta,s}$	$\delta_{1,s}$	$\sigma_{\varepsilon,s}$
S0	t; parallel trends	0	0	1
S1	t; weak heterogeneity	0.0200	10	1
S2	t; moderate heterogeneity	0.0500	10	1
S3	t; strong heterogeneity	0.1000	10	1
S4	t^2; moderate heterogeneity	0.0025	200	1
S5	t^2; strong heterogeneity	0.0050	200	1
S6	t; moderate heterogeneity, high noise	0.0500	10	3

$$N = 200, \quad T = 20, \quad T_0 = 12, \quad \tau = 2, \quad R = 500.$$

All six estimators use the same panel within each replication: 3,500 panels in total.

Simulation Forecast Specifications

Estimator	Implemented specification
RF adj.	500 trees, 2 candidate predictors per split, terminal node size 5.
GB adj.	800 trees, depth 3, shrinkage 0.01, minimum node size 10, bag fraction 0.7.
SC adj.	Nonnegative donor weights sum to one; the sole matching predictor is the pre-treatment mean.

We use conventional model specifications with fixed settings throughout the simulation.

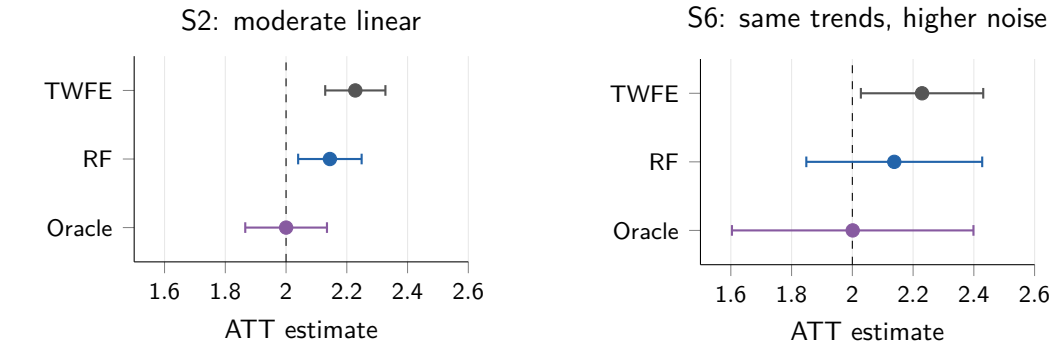
Bias Reduction in ATT Estimates

Signed Monte Carlo bias: $R^{-1}\sum_r(\widehat{\tau}_r - 2)$. Bold: smallest absolute bias among non-oracle methods.

Scenario	TWFE	Oracle	Linear	SC	RF	GB
S0 (parallel trends)	-0.0058	-0.0058	-0.0058	-0.0046	**-0.0014**	-0.0034
S1 (weak linear heterogeneity)	0.0406	-0.0035	0.0406	0.0413	**0.0273**	0.0476
S2 (moderate linear heterogeneity)	0.2280	0.0002	0.2280	0.2221	**0.1443**	0.2522
S3 (strong linear heterogeneity)	0.8222	-0.0045	0.8222	0.6914	**0.4942**	0.6997
S4 (moderate quadratic heterogeneity)	0.2647	-0.0314	0.2647	0.2652	**0.1918**	0.2747
S5 (strong quadratic heterogeneity)	0.9194	-0.0023	0.9194	0.8927	**0.6692**	0.8761
S6 (moderate linear heterogeneity, high noise)	0.2295	0.0012	0.2295	0.2702	**0.1380**	0.2472

- Stronger trend differences create larger TWFE bias.
- RF has the smallest absolute bias among non-oracle methods in every scenario.
- Oracle is closest to zero in S1–S6, but lower bias alone does not guarantee a better estimate.

Lower Bias Does Not Always Mean More Stable Estimates



Dots: mean estimates. Lines: mean $\pm$ 1 SD. Dashed line: true ATT $= 2$.

- RF and Oracle still move the average estimate toward the truth.
- Higher noise makes individual estimates much more variable.

We need to consider both bias and dispersion.

Overall Estimation Accuracy Relative to TWFE

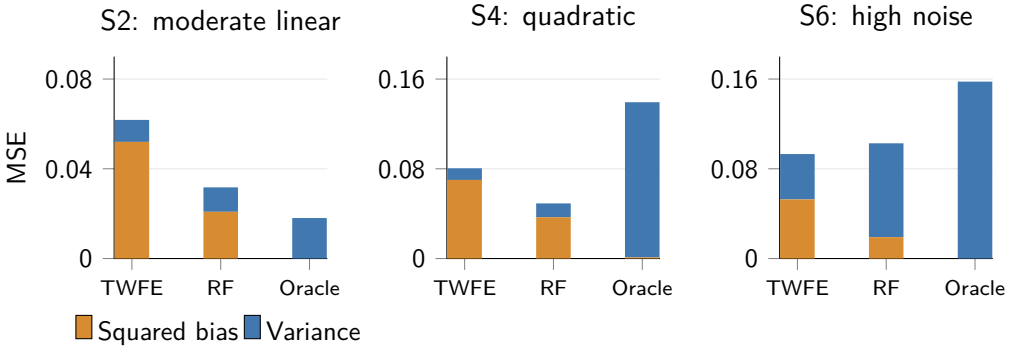
$\mathrm{RMSE} = \{R^{-1}\sum_r(\widehat{\tau}_r - 2)^2\}^{1/2}$. Bold: lowest RMSE among non-oracle methods.

Scenario	TWFE	Oracle	Linear adj.	SC adj.	RF adj.	GB adj.
S0	0.0651	0.0651	0.0651	**0.0645**	0.0965	0.0787
S1	**0.0819**	0.1372	**0.0819**	0.0822	0.0988	0.1024
S2	0.2486	0.1344	0.2486	0.2428	**0.1781**	0.2736
S3	0.8345	0.1377	0.8345	0.7089	**0.5088**	0.7093
S4	0.2836	0.3733	0.2836	0.2841	**0.2217**	0.2972
S5	0.9314	0.4058	0.9314	0.9054	**0.6827**	0.8860
S6	**0.3052**	0.3971	**0.3052**	0.3338	0.3205	0.3184

- RF lowers RMSE by 28.4%, 39.0%, 21.8% and 26.7% in S2–S5.
- GB improves RMSE in S3 and S5; SC gains depend on scenario.

Bias Reduction and Overall Accuracy

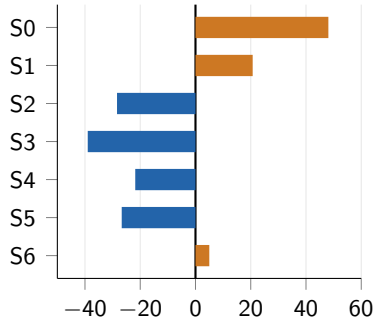
$$\mathrm{MSE} = \mathrm{Bias}^2 + V_R$$



V_R uses denominator R. Bars use rounded table entries; panel scales differ.

Oracle reduces average bias, but extra variation can outweigh the gain.

When Is Correction Worth Using?



Negative: lower RMSE. Positive: higher RMSE.

S0–S1: parallel trends or weak divergence

RF RMSE rises by 48.1% and 20.7%.
Keep no correction as a serious candidate.

S2–S5: larger divergence, lower noise

RF RMSE falls by 21.8–39.0%.
Correction can deliver meaningful gains.

S6: higher noise

RF lowers bias, but RMSE rises by 5.0%.
Check whether the bias gain survives the variance cost.

Ask whether correction helps at all, as well as which method to use.

How Should We Choose a Method in Practice?

In simulation	In an application
The true effect is known: $\tau = 2$.	The true effect is unknown.
We measure ATT estimation error directly.	We cannot directly rank true ATT errors.
Oracle uses the known scenario form.	The untreated trend form is unknown.

How can we compare methods before estimating the effect?

1. Common benchmark

Compare corrections with the same DID estimate, including no correction.

2. Pre-treatment tests

Use earlier untreated periods to evaluate the forecasting task.

3. Select, then estimate

Choose using pre-treatment performance, then estimate the intervention contrast.

Simulation motivates the comparison; pre-treatment tests make the choice operational.
Historical performance informs the choice, but does not reveal the missing counterfactual.

Cear'a: Police Withdrawal and Gang Exposure

André Mancha (2025)
When the state steps down: Reduced police surveillance and gang-related deaths in Brazil

The event and comparison

- Statewide police strike: 18 February–1 March 2020.
- Compare high versus lower gang exposure.
- 84 pre-strike days; 13 strike days.

The data and groups

- Daily homicides in 358 municipality/district units, within 22 AIS.
- 5 high-exposure AIS: 94 treated units; 264 controls.

Municipality/district: the unit observed each day.

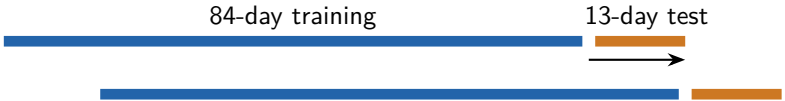
AIS: the larger security area used to define exposure groups and cluster inference.

What we estimate

Both groups experienced the strike. We estimate the change in the high-minus-lower-exposure gap relative to its predicted no-strike path.

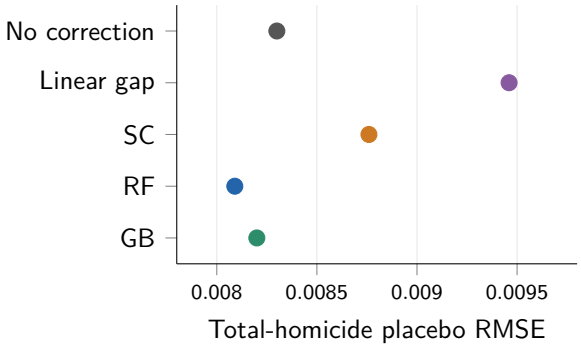
Testing the Methods Before the Strike

The true strike-period effect is unknown, so we test the methods in earlier periods.



1. Fit each method using **84 days**.
2. Predict the following **13 days**.
3. Move forward by **14 days** and repeat across **154 windows**.
4. Select the lowest placebo RMSE **before examining strike-period estimates**.

RF Has the Lowest Pre-Strike RMSE, with a Small Gain

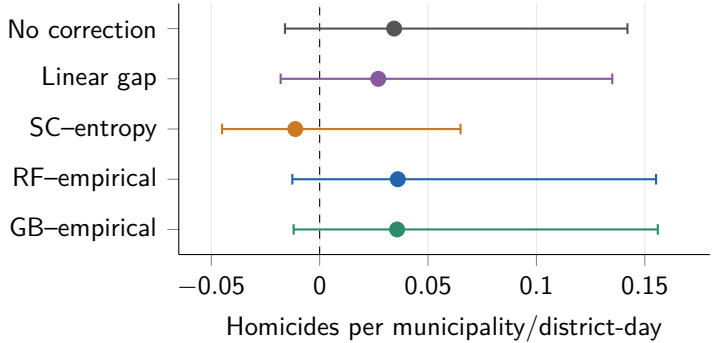


RF: 0.00809

No correction: 0.00830

- RF is selected by the pre-strike RMSE rule.
- Its gain over no correction is modest: 2.6%.
- No correction remains competitive; Linear gap and SC have higher RMSE.

Strike-Period Estimates



Dots: point estimates. Lines: 95% bootstrap intervals.

Selected RF:
0.03606

TWFE: 0.03437

- RF remains close to TWFE.
- SC is negative; the other estimates are positive.

Sensitivity to Treatment Definition

Total-homicide comparison	Estimate or magnitude
Five-AIS TWFE, main definition	0.03437
Four-AIS TWFE, executed source-code definition	0.05677
Change in TWFE	≈ 0.0224
Magnitude of the selected RF correction	0.00169

- This treatment-definition change is about 13 times the RF adjustment.
- Main analysis follows the paper's intended five-AIS definition.
- A separate AIS 07 cutoff issue motivates the six-AIS sensitivity.

What This Framework Adds—and Where It Stops

Contributions

1. **Reframe the DID problem.**
   Forecast and correct a changing untreated group gap.
2. **Provide a transparent workflow.**
   Subtract the predicted gap change from DID; keep the benchmark and weights consistent.
3. **Show how to evaluate correction.**
   Use known effects in simulation and pre-treatment tests in the application.

Limits

1. **The counterfactual needs an assumption.**
   Good earlier forecasts do not guarantee the no-strike path.
2. **The simulation covers a specific setting.**
   One treatment date, constant effects, independent errors and fixed settings.
3. **The causal contrast is limited.**
   Both groups experienced the strike; only five treated AIS limit inference.

Conclusions

1. Forecasting can improve DID when untreated trends differ across groups.
2. In S2–S5, RF has the lowest RMSE among non-oracle methods, reducing RMSE by about **22–39%** relative to TWFE.
3. Lower bias does not always mean greater accuracy: the gain can be offset by higher variance.

Forecast the gap, correct DID, and evaluate the full estimation error.